Mastering Game Bots

Lecture 1



- Course rules
- CodinGame
- Discussing the games (past contest)

Course lectures (plan)

Lecture 1 (07.10): Introduction

Introduction. CodinGame. Game types. Discussing the games from past contests – project proposals.

Lecture 2 (14.10): Basics 1

Local benchmarker setup. Game state. Evaluation function (ingredients, examples).

Lecture 3 (21.10): Basics 2

Greedy strategy (Hill Climbing). Iterative opponent prediction. Confidence, sequential testing. Versionization, paper-rock-scissors syndrome. Tuning weights.

Lecture 4 (28.10, 12:15): AI, LLMs

Guest lecture by Daniel Górski (Zylo) – Automatic bot development for CodinGame.

Followed by open lecture at 14:15 (room 119) – Orchestrating AI.

Lecture 5 (18.11): Algorithms

- **Beam Search**

Straight vs nested, limiter, greedy strategy, predicting opponent, action pruning, time management. Efficient implementation.

- **MCTS**

Decoupled, UCT forest. Efficient implementation. Light vs heavy simulation, fixed policy in rollout, action pruning.

Lecture 6 (25.11): Engine

Making pixel-perfect engine; *reveal* technique. Debugging, time management, measuring performance, timed vs fixed. Advanced optimizations (preprocessing, engine rewriting, bitboarding, AVX).

Lecture 7 (02.12): Cultist Wars

Mastering the Cultist Wars – an example of #1 bot. (Minimax-based.)

The timeline is a bit flexible. The lectures may be shifted when needed (in particular, if a contest happens in the meantime).

Goal

- I want to show you overview of techniques and my pipeline developed during the last few years.
- You are expected to participate in future contests – we need to build a new team.

Required knowledge:

- Most common search algorithms (e.g., MCTS, Beam Search, Minimax).
- C++.
- Java (may be useful for game referees).

There will be follow-up course ``Project: Mastering Game Bots`` in the summer semester 2027. You can continue development there or choose another game to master.

Course laboratory

Laboratory

- You choose one bot programming game.
- Assignment tasks drive progressive development of the bot.
- The bot becomes finally your ***project*** – continue improving to place as high as possible on the leaderboard.
  The deadline is the last lab before the session (3th February).

Grading

>= 100% – 5.0 and exam exemption with a 5.0 mark
90% – 5.0 and exam exemption with a 4.5 mark
80% – 4.5 and exam exemption with a 4.0 mark
70% – 4.0
60% – 3.5
50% – 3.0

Course exam

- The exam will take place during the session (after evaluating the projects).
- Written exam with open and closed questions.
- About 1h.
- Exemptions are obtained by good laboratory results.

CodinGame technical rules

- Bots must be prepared individually — no code sharing or publishing.
- No multi accounts.
- Time limits (most common): 50 ms per move; 1 second for the first move.
- Memory limit: 768 MiB (using less is recommended for faster execution).
- The code is limited to 100,000 characters

  (UTF-16, however;

  `ce static UTF16 E=L"徨⊗怯ią褫ώ眛悅胼忿の惚ど庇□怍□,囲ゝ徴□恊褄恃鶉忼□"`).
- It is forbidden to carry and run binary code encoded in the submission.
- Manual plays and submissions are limited in a period of time

  (e.g., 80 plays/1h; 600 plays/24h).
- LLMs allowed.

Technical recommendations

Recommended software:

- Linux.
- GCC (g++).
- IntelliJ (for Java Benchmarker and referee) – the free version is sufficient.
- CLion or Visual Studio Code (for the bot).

Recommended hardware:

- Many CPU cores.

LLM policy

- LLMs are allowed as you like, but you need to understand your code for the purposes of this course.
  (However, you do not need to understand your code for a real contest).

Recommended way of using LLMs:

- Reviewing the code; finding bugs.
- Generating ideas (e.g., how to optimize, new ingredients in evaluation).
- Generating specific well-defined functions.
- Generating AVX code.
- Generating auxiliary scripts (Python).

(Still) not recommended:

- Generating the whole code.
- Generating efficiency-crucial parts (algorithms, data structures, game engine).
- Generating long pieces of code.

Other programming contests

AtCoder 2026

Customize

Rank	User		Score	A
1	OpenAI	OpenAI	50,000,000,000 (25) 590:08	50,000,000,000 (25) 590:08
2	terry_u16	株式会社 ALGO ARTIS	3,081,190,090 (13) 583:47	3,081,190,090 (13) 583:47
3	Shun_PI	株式会社THIRD	2,313,596,445 (10) 593:00	2,313,596,445 (10) 593:00
4	yosupo	Indeed / Recruit	1,909,318,961 (7) 516:24	1,909,318,961 (7) 516:24
5	yokozuna57	Preferred Networks, Inc.	601,626,335 (7) 597:03	601,626,335 (7) 597:03
6	asi1024	Preferred Networks, Inc.	464,877,204 (6) 530:59	464,877,204 (6) 530:59
7	montplusa	フューチャー株式会社	410,528,989 (10) 523:19	410,528,989 (10) 523:19
8	nikaj		321,945,361 (2) 611:48	321,945,361 (2) 611:48
9	saharan	????????? ????????, ???	231,038,286 (1) 391:25	231,038,286 (1) 391:25
10	rhoo	Institute of Science Tokyo	228,270,028 (1) 170:45	228,270,028 (1) 170:45
11	bowwowforeach	株式会社THIRD	37,832,289 (3) 598:40	37,832,289 (3) 598:40
12	eijirou	株式会社 ALGO ARTIS	2,314,199 230:29	2,314,199 230:29
13	MathGorilla	株式会社 ALGO ARTIS	2,156,581 195:37	2,156,581 195:37
14	Rafbill		1493 (4) 447:42	1493 (4) 447:42

ICPC 2025

OpenAI has solved all 12 problems at ICPC 2025, beating all humans.

CodinGame

Top contest players at CodinGame wrote (at least some) manual code at recent contests.

Contest overview

- Most top bots for contest games:
  - are in C++
  - use simulation
  - or use Reinforcement Learning
    - for action selection (PPO)
    - for evaluation
- Postmortems (forum)
- Most popular search algorithms:
  - Hill climbing (HC)
  - MCTS + UCT
  - Best first search
  - Minimax
  - Beam search
  - Iterative greedy

Game types

- Perfect information vs imperfect information
- Alternate vs simultaneous moves
- With randomness or without
- Random initial board or fixed

CodinGame – past contest games

- Green Circle (2022)
- Cultist Wars (2022)
- Keep on the Grass (2022) – Fall Challenge
- Fourmis (Ants) (2023) – Spring Challenge
- Seabed Security (2023) – Fall Challenge
- Olymbits (2024) – Spring Challenge
- Cellularena (2024) – Winter Challenge
- Soak Overflow (2025) – Summer Challenge
- SnakeByte (2026) – Winter Challenge
- Troll Farm (2026) – Spring Challenge

Assignment 1

01 Select a Game
Choose one of the bot programming games.
(Past contest games strongly recommended, and preferably, a game you have not worked before.)

02 Study Strategies
Read Post-Mortems (PM) of top players on the CodinGame forum.

03 Reach Bronze League
Advance at least to the Bronze League to unlock and access full rules.

04 Consult & Discuss
Discuss the game and your ideas with me (can be online via Discord).

05 Submit Information
Submit the link to your CodinGame profile and the link to game at SKOS as a solution to Assignment 1.
Ensure you have “University of Wrocław” as the school in your profile.



DEADLINE
October 21

Enroll in the course at SKOS – it is temporarily open.